

* Syntax-:

`int l, r; a = new int[5]`

* Count greater than itself-:

`a = [ -3, -2, 6, 8, 4, 8, 5 ]`

ans = 5

* Observation-:

→ for every max element there won't be any element greater than it

→ for every other element, there will be at most one element greater than it

1) Iterate & get max value

2) Iterate & get count of elements $! = \text{max}$

Tc: $O(N)$

Sc: $O(1)$

* Check Pairs (i, j) :-

Such that

$$a[i] + a[j] = k \quad \& \quad i \neq j$$

$a = [2, 3, -2, 1, 4, 5, 6, 8]$

$k = 10$

$$4 + 6 = 10 \quad \text{true}$$

Idea:-

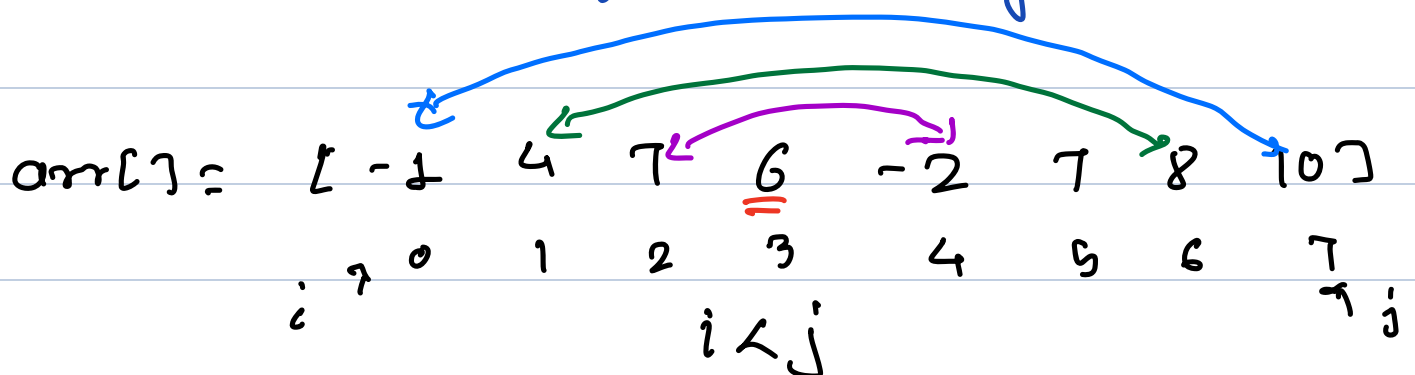
→ Run loop with i for all indices

→ Run a nested loop with j over all indices.

$$T.C: O(N^2)$$

$$S.C: O(1)$$

* Reverse entire Array:-



$$T.C: O(\log N)$$

$$S.C: O(1)$$

* Reverse part of Array (S to E)

OL: [-3 4 2 8 7 9 6 2 10]
0 1 2 3 4 5 6 7 8

[S E] = [3 7]

TC: $O(N)$

SC: $O(1)$

* Rotate Array from last to first
by K times-: (Note-: $K > 0$)

$K = 5$

$a[13] = [a_0, a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9, a_{10}, a_{11}, a_{12}]$

Reverse the entire array

$[a_{12}, a_{11}, a_{10}, a_9, a_8, a_7, a_6, a_5, a_4, a_3, a_2, a_1, a_0]$

reverse the first K elements Reverse the last $(N-K)$ elements

rotate by 5 times-:

$a_8, a_9, a_{10}, a_{11}, a_{12}, a_0, a_1, a_2, a_3, a_4, a_5, a_6, a_7$

1) Reverse entire array, $S=0, E=N-1$

2) Reverse first K elements, $S=0, E=K-1$

3) Reverse last $N-k$ elements, $S=k$

$$C = N-1$$

if $k > N$

$$k = k \% N$$

TC: $O(N)$

SC: $O(1)$